Exp No: 01

STUDY OF NETWORKING BASIC COMMANDS IN LINUX AND WINDOWS

AIM:

To study and execute various networking commands in Linux and Windows to understand network resolution and packet analysis.

COMMANDS:

1) hostname:

Display the hostname of the system.

2) ipconfig/all (window)

Display complete TCP/IP configuration details.

3) arp -a

Displays the ARP table containing IP and MAC addresses.

4) Netstat -r

Display routing table and active network connections.

5) nslookup (www.google.com)

Retrieves DNS information of a domain.

6) Ping (google.com)

Check network connectivity between the local machine and destination host.

7) ip (address show) linux commands.

Displays ip address configuration of network interfaces.

8) if config

Display network interface configuration.

9) mtr (google.com)

Shows packet route and packet loss statistics.

```
Microsoft Windows [Version 10.0.26200.8737]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Ramya>arp -a

Interface: 192.168.56.1 --- 0xc
  Internet Address      Physical Address      Type
  192.168.56.255        ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.255.250       01-00-5e-7f-ff-fa     static

Interface: 192.168.0.135 --- 0x10
  Internet Address      Physical Address      Type
  192.168.0.1           e0-1c-fc-e5-47-c6     dynamic
  192.168.0.27          ca-7b-57-b2-7d-22     dynamic
  192.168.0.71          80-47-86-63-57-c8     dynamic
  192.168.0.255         ff-ff-ff-ff-ff-ff     static
  224.0.0.2             01-00-5e-00-00-02     static
  224.0.0.22            01-00-5e-00-00-16     static
  224.0.0.251           01-00-5e-00-00-fb     static
  224.0.0.252           01-00-5e-00-00-fc     static
  239.255.102.18        01-00-5e-7f-66-12     static
  239.255.255.250       01-00-5e-7f-ff-fa     static
  255.255.255.255       ff-ff-ff-ff-ff-ff     static
```

```
C:\Users\Ramya>hostname
DESKTOP-0NRMEE7
```

```
C:\Users\Ramya>netstat -r
===========================================================================
Interface List
 12...0a 00 27 00 00 0c ......VirtualBox Host-Only Ethernet Adapter
  9...7c 2a 31 e4 02 10 ......Microsoft Wi-Fi Direct Virtual Adapter
  6...7e 2a 31 e4 02 0f ......Microsoft Wi-Fi Direct Virtual Adapter #2
 16...7c 2a 31 e4 02 0f ......Intel(R) Dual Band Wireless-AC 8265
  1...........................Software Loopback Interface 1
===========================================================================

IPv4 Route Table
===========================================================================
Active Routes:
Network Destination        Netmask          Gateway       Interface  Metric
          0.0.0.0          0.0.0.0      192.168.0.1    192.168.0.135     40
        127.0.0.0        255.0.0.0         On-link         127.0.0.1    331
        127.0.0.1  255.255.255.255         On-link         127.0.0.1    331
  127.255.255.255  255.255.255.255         On-link         127.0.0.1    331
      192.168.0.0    255.255.255.0         On-link     192.168.0.135    296
    192.168.0.135  255.255.255.255         On-link     192.168.0.135    296
    192.168.0.255  255.255.255.255         On-link     192.168.0.135    296
     192.168.56.0    255.255.255.0         On-link      192.168.56.1    281
     192.168.56.1  255.255.255.255         On-link      192.168.56.1    281
   192.168.56.255  255.255.255.255         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link         127.0.0.1    331
        224.0.0.0        240.0.0.0         On-link      192.168.56.1    281
        224.0.0.0        240.0.0.0         On-link     192.168.0.135    296
  255.255.255.255  255.255.255.255         On-link         127.0.0.1    331
  255.255.255.255  255.255.255.255         On-link      192.168.56.1    281
  255.255.255.255  255.255.255.255         On-link     192.168.0.135    296
===========================================================================
Persistent Routes:
  None
```

```
C:\Users\Ramya>nslookup www.google.com
Server:  UnKnown
Address:  fe80::e21c:fcff:fee5:47c6

Non-authoritative answer:
Name:    www.google.com
Addresses:  2001:4860:482d:7700::
          2001:4860:4826:7700::
          2001:4860:482a:7700::
          2001:4860:4829:7700::
          2001:4860:4827:7700::
          2001:4860:4828:7700::
          2001:4860:482b:7700::
          2001:4860:482c:7700::
          142.251.156.119
          142.251.157.119
          142.251.150.119
          142.251.155.119
          142.251.152.119
          142.251.153.119
          142.251.151.119
          142.251.154.119
```

10.) tcpdump (sudo tcpdump -i eth 0 -c 10)

Captures and displays network packets.

11.) nmcli (nmcli connection show)

Displays available network connections profiles.

12) ping :

Syntax : ping localhost

13) Route :

Displays the IP routing table.

RESULT:

Thus the various networking commands in linux and windows were successfully executed and studied.

```
C:\Users\Ramya>ping google.com

Pinging google.com [142.250.205.110] with 32 bytes of data:
Reply from 142.250.205.110: bytes=32 time=4ms TTL=118
Reply from 142.250.205.110: bytes=32 time=3198ms TTL=118
Reply from 142.250.205.110: bytes=32 time=3ms TTL=118
Reply from 142.250.205.110: bytes=32 time=6ms TTL=118

Ping statistics for 142.250.205.110:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 3198ms, Average = 802ms
```

```
C:\Users\Ramya>ping

Usage: ping [-t] [-a] [-n count] [-l size] [-f] [-i TTL] [-v TOS]
            [-r count] [-s count] [[-j host-list] | [-k host-list]]
            [-w timeout] [-R] [-S srcaddr] [-c compartment] [-p]
            [-4] [-6] target_name

Options:
    -t                 Ping the specified host until stopped.
                       To see statistics and continue - type Control-Break;
                       To stop - type Control-C.
    -a                 Resolve addresses to hostnames.
    -n count           Number of echo requests to send.
    -l size            Send buffer size.
    -f                 Set Don't Fragment flag in packet (IPv4-only)
    -i TTL             Time To Live.
    -v TOS             Type Of Service (IPv4-only. This setting has been deprecated
                       and has no effect on the type of service field in the IP
                       Header).
    -r count           Record route for count hops (IPv4-only).
    -s count           Timestamp for count hops (IPv4-only).
    -j host-list       Loose source route along host-list (IPv4-only).
    -k host-list       Strict source route along host-list (IPv4-only).
    -w timeout         Timeout in milliseconds to wait for each reply.
    -R                 Use routing header to test reverse route also (IPv6-only).
                       Per RFC 5095 the use of this routing header has been
                       deprecated. Some systems may drop echo requests if
                       this header is used.
    -S srcaddr         Source address to use.
    -c compartment     Routing compartment identifier.
    -p                 Ping a Hyper-V Network Virtualization provider address.
    -4                 Force using IPv4.
    -6                 Force using IPv6.
```

```
C:\Users\Ramya>ipconfig

Windows IP Configuration


Ethernet adapter Ethernet:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::3716:46e4:5f1a:704b%12
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 10:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : fd01::3bdb:596d:9209:8711
   Temporary IPv6 Address. . . . . . : fd01::31fb:78ab:4a7f:5820
   Link-local IPv6 Address . . . . . : fe80::a4a1:9676:dd78:c212%16
   IPv4 Address. . . . . . . . . . . : 192.168.0.135
   Subnet M[illegible]
   Default [illegible]
```

```
C:\Users\Ramya>route

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f               Clears the routing tables of all gateway entries.  If this is
                   used in conjunction with one of the commands, the tables are
                   cleared prior to running the command.

  -p               When used with the ADD command, makes a route persistent across
                   boots of the system. By default, routes are not preserved
                   when the system is restarted. Ignored for all other commands,
                   which always affect the appropriate persistent routes.

  -4               Force using IPv4.

  -6               Force using IPv6.

  command          One of these:
                     PRINT     Prints  a route
                     ADD       Adds    a route
                     DELETE    Deletes a route
                     CHANGE    Modifies an existing route
  destination      Specifies the host.
  MASK             Specifies that the next parameter is the 'netmask' value.
  netmask          Specifies a subnet mask value for this route entry.
                   If not specified, it defaults to 255.255.255.255.
  gateway          Specifies gateway.
  interface        the interface number for the specified route.
  METRIC           specifies the metric, ie. cost for the destination.

All symbolic names used for destination are looked up in the network database
file NETWORKS. The symbolic names for gateway
```